Ps 06

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1 Problem 1

1.1 Task

(... used in the proof of the Taylor formula)

(a) Suppose that $\alpha \in \mathbb{N}^n$ and $h \in \mathbb{R}^n$ such that $\|h\|_\infty \leq 1$. Prove that

$$|h^\alpha| \leq \|h\|_\infty^{|\alpha|}$$

(b) Let $k \in \mathbb{N}$ and suppose that $f : \mathbb{R}^n \to \mathbb{R}$ is k times continuously differentiable. For $x, x_0 \in \mathbb{R}^n$ and $h = x - x_0$, define

$$g : [-1, 1] \to \mathbb{R}, \quad g(t) := f(x_0 + th)$$

For $j \in \{0, \ldots, k\}$ prove that g is k times continuously differentiable and

$$g^{(j)}(t) = \sum_{\substack{\alpha \in \mathbb{N}^n \\ |\alpha| = j}} \frac{j!}{\alpha!} (\partial^\alpha f)(x_0 + th)\, h^\alpha, \quad t \in [-1, 1]$$

(hint: induction and chain rule)

1.2 Solution

1.2.1 (a)

Define

$$|h^\alpha| = \prod_{i=1}^n |h_i|^{\alpha_i}, \quad \|h\|_\infty^{|\alpha|} = \prod_{i=1}^n \|h\|_\infty^{\alpha_i}$$

As, $|h_i| \leq \|h\|_\infty$,

$$|h^\alpha| = \prod_{i=1}^n |h_i|^{\alpha_i} \leq \prod_{i=1}^n \|h\|_\infty^{\alpha_i} = \|h\|_\infty^{|\alpha|}$$

1.2.2 (b)

Setup. For $k \in \mathbb{N}$, $f : \mathbb{R}^n \to \mathbb{R}$, $f \in C^k$. Let $x, x_0 \in \mathbb{R}^n$, $h = x - x_0$. Define $g : [-1, 1] \to \mathbb{R}$, $g(t) := f(x_0 + th)$.

Proposition. for $j \in \{0, \ldots, k\}$:

1. $g \in C^k$
2. For such j, $g^{(j)}(t) = \sum_{|\alpha|=j} \frac{j!}{\alpha!}(\partial^\alpha f)(x_0 + th)\, h^\alpha$

Base case. $j = 0$, only $\alpha = 0 \in \mathbb{N}^n$ contributes.

$$\begin{aligned} g^{(0)}(t) &= \sum_{|\alpha|=0} \frac{0!}{0!\dots 0!}(\partial_1^0 \dots \partial_n^0 f)(x_0 + th) h_1^0 \dots h_n^0 \\ &= \frac{1}{1} f(x_0 + th) \cdot 1 \\ &= f(x_0 + th) \end{aligned}$$

Holds.

Inductive hypothesis.

2 Problem 2

2.1 Task

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable.

(a) Prove that the Taylor formula centered at $x_0 \in \mathbb{R}^n$ of order 2 is

$$f(x) = f(x_0) + (\nabla f(x_0))^\top (x - x_0) + \frac{1}{2}(x - x_0)^\top H_f(x_0)(x - x_0) + o(\|x - x_0\|^2)$$

(b) Implement and plot the Taylor approximation of f of order 2 by a heatmap for the function

$$f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = x^4 + y^4 + x^2 + y^2$$

2.2 Solution

2.2.1 (a)

$$f(x) = \sum_{|\alpha| \le 2} \frac{\partial^\alpha f(x_0)}{\alpha!}(x - x_0)^\alpha + o(\|x - x_0\|^2) \quad (x \to x_0)$$

3 Problem 3

3.1 Task

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$,

$$f(x, y) = x^2 + y^2 + 2x + 4y$$

(a) Determine all local extrema of f.

(b) Plot f in Julia by a heatmap to check your calculations in Part (a).

3.2 Solution

3.2.1 (a)

The function f is twice continuously differentiable on $\mathbb{R}^2$. Compute the partial derivatives:

$$\partial_1 f(x, y) = 2x + 2, \quad \partial_2 f(x, y) = 2y + 4.$$

A critical point satisfies $\operatorname{grad} f(x, y) = 0$, i.e.

$$\begin{aligned} 2x + 2 = 0 &\implies x = -1, \\ 2y + 4 = 0 &\implies y = -2. \end{aligned}$$

Hence f has the unique critical point $x_0 = (-1, -2)$.

The second-order partial derivatives are

$$\partial_1 \partial_1 f = 2, \quad \partial_2 \partial_2 f = 2, \quad \partial_1 \partial_2 f = \partial_2 \partial_1 f = 0,$$

so the Hessian matrix at x_0 is

$$H_f(x_0) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

For any $h = (h_1, h_2) \in \mathbb{R}^2 \setminus \{0\}$,

$$h^\top H_f(x_0) h = 2h_1^2 + 2h_2^2 > 0,$$

so $H_f(x_0)$ is positive definite. By Theorem 7.2.5(1), $x_0 = (-1, -2)$ is a strict local minimum, with

$$f(-1, -2) = (-1)^2 + (-2)^2 + 2(-1) + 4(-2) = -5.$$

Since x_0 is the only critical point and H_f is positive definite on all of $\mathbb{R}^2$, f has no other local extrema.

3.2.2 (b)

```
using Plots

f(x, y) = x^2 + y^2 + 2x + 4y

xs = range(-5, 3, length=200)
ys = range(-6, 2, length=200)

heatmap(xs, ys, (x, y) -> f(x, y), aspect_ratio=1,
        xlabel="x", ylabel="y", title="f(x,y) = x² + y² + 2x + 4y")

scatter!([-1], [-2], color=:red, label="min (-1,-2)")

mkpath("./assets")
savefig("./assets/p_03_b.png")
```

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/Users/ziro2173/personal/mfds/01_bachelor/02_semester/01_analysis/02_exercises/ps_06/assets/p_03
```

4 Problem 4

4.1 Task

Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$,

$$f(x, y) = x^2 + 3y^2 + 2xy$$

(a) Determine all local extrema of f.

(b) Plot f in Julia by a heatmap to check your calculations in Part (a).

4.2 Solution

4.2.1 (a)

The function f is twice continuously differentiable on $\mathbb{R}^2$. Compute the partial derivatives:

$$\partial_1 f(x,y) = 2x + 2y, \quad \partial_2 f(x,y) = 6y + 2x.$$

A critical point satisfies $\operatorname{grad} f(x,y) = 0$, i.e.

From the system:

$$\begin{aligned} 2x + 2y = 0 &\implies x = -y, \\ 6y + 2x = 0 &\implies x = -3y. \end{aligned}$$

Combining gives $-y = -3y$, hence $y = 0$ and $x = 0$. Thus f has the unique critical point $x_0 = (0,0)$.

The second-order partial derivatives are

$$\partial_1\partial_1 f = 2, \quad \partial_2\partial_2 f = 6, \quad \partial_1\partial_2 f = \partial_2\partial_1 f = 2,$$

so the Hessian matrix at x_0 is

$$H_f(x_0) = \begin{pmatrix} 2 & 2 \\ 2 & 6 \end{pmatrix}.$$

For any $h = (h_1, h_2) \in \mathbb{R}^2 \setminus \{0\}$,

$$\begin{aligned} h^\top H_f(x_0) h &= 2h_1^2 + 4h_1h_2 + 6h_2^2 \\ &= 2(h_1^2 + 2h_1h_2 + h_2^2) + 4h_2^2 \\ &= 2(h_1 + h_2)^2 + 4h_2^2 > 0, \end{aligned}$$

since the sum vanishes only if $h_2 = 0$ and $h_1 + h_2 = 0$, i.e. $h = 0$. Hence $H_f(x_0)$ is positive definite. By Theorem 7.2.5(1), $x_0 = (0,0)$ is a strict local minimum, with

$$f(0,0) = 0.$$

Since x_0 is the only critical point and H_f is constant and positive definite on all of $\mathbb{R}^2$, f has no other local extrema.

4.2.2 (b)

```
using Plots

f(x, y) = x^2 + 3y^2 + 2x*y

xs = range(-4, 4, length=200)
ys = range(-4, 4, length=200)

heatmap(xs, ys, (x, y) -> f(x, y), aspect_ratio=1,
        xlabel="x", ylabel="y", title="f(x,y) = x² + 3y² + 2xy")

scatter!([0], [0], color=:red, label="min (0,0)")

mkpath("./assets")
savefig("./assets/p_04_b.png")
```

```
/Users/ziro2173/personal/mfds/01_bachelor/02_semester/01_analysis/02_exercises/ps_06/assets/p_04
```

5 Errors/Typos

- Problem 1(a): "$\|h\|^{|\alpha|}$" $\to$ "$\|h\|_\infty^{|\alpha|}$" (missing ∞ subscript on RHS; the natural bound from $|h_i| \leq \|h\|_\infty$ is in $\|h\|_\infty$)
- Problem 1(b): "$x, x_0 \in \mathbb{R}$" $\to$ "$x, x_0 \in \mathbb{R}^n$" (required since $h = x - x_0 \in \mathbb{R}^n$)
- Problem 2(a): "$x_0 \in \mathbb{R}$" $\to$ "$x_0 \in \mathbb{R}^n$"
- Problem 2(a): "$(\nabla f(x_0))^\top (x - x_0)^\alpha$" $\to$ "$(\nabla f(x_0))^\top (x - x_0)$" (extraneous α exponent; no multi-index α is defined in this problem)